Kartik Loya

Ph.D. in Mechanical Engineering

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Summary

Ph.D. in Mechanical Engineering with hands-on experience in robotics, dynamics, perception, and control. I build and test robotic systems, combining sensing, physics-based modeling, and learning-based methods to develop reliable autonomy for real-world environments. My work spans system identification, state estimation, planning, and real-time control, with a growing interest in Physical AI, reinforcement learning, and learning-based approaches for intelligent robotic behavior.

Education

Clemson University, Ph.D. in Mechanical Engineering – Clemson, SC, USA 2021-08 - 2026-12

- GPA: 4.00/4.00

Clemson University, M.S. in Mechanical Engineering – Clemson, SC, USA Aug 2018 – Aug 2021

- GPA: 3.87/4.00.

Maharashtra Institute of Technology, Pune University, B.E. in Mechanical Engineering – Pune, India Aug 2014 – May 2018

Skills

Programming: Python, MATLAB, C++

Robotics & Machine Learning: ROS, ROS2, PyTorch, OpenCV, MuJoCo

Simulation & Optimization: Project Chrono, CasADi, IPOPT, GPOPS-II, SolidWorks

Embedded Computing: Linux, Git, Docker, Raspberry Pi, Radxa, NVIDIA Jetson Nano, ESP32

Sensors & Perception: IMU, GPS, LiDAR, RGB/depth cameras, Intel RealSense, pressure sensors, wheel encoders

Actuation & Communication: BLDC motors, servo motors, motor drivers, encoder feedback, CAN, UART, I2C, SPI, RF communication

Research Experience

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA Sept 2024 – present

Advisor: Prof. Phanindra Tallapragada

- Developed a physics-informed machine learning framework for off-road autonomy on deformable terrain, learning fast vehicle-dynamics predictors from large-scale simulation data for real-time control.
- Built terrain-specific dynamics models using scalable Koopman identification, demonstrating strong performance degradation across unseen soil conditions and motivating terrain-adaptive learning and control.
- Integrated learned dynamics with constrained MPC for aggressive trajectory tracking under steering and torque limits, achieving real-time control at 0.085s per optimization step.
- Currently deploying the framework on a 1/10th-scale autonomous off-road vehicle, integrating depth camera, wheel encoders, IMU, and GPS for onboard perception, state estimation, terrain-aware modeling, and closed-loop control.

Graduate Research Assistant, Applied Dynamical Systems Lab – Clemson, SC, USA Aug 2021 – May 2024

Advisor: Prof. Phanindra Tallapragada

- Designed, built, and field-tested a fish-like underwater robot in pool and lake environments, integrating BLDC internal-rotor propulsion, servo-actuated flexible tail, IMU, pressure sensors, GPS, onboard compute, and overhead vision-based marker tracking for localization and state estimation.
- Developed a physics-based nonlinear simulator capturing hydrodynamics, underactuation, and coupled body–tail dynamics for scalable data generation, system identification, and controller development.
- Studied roll stability of the bio-inspired fish robot, showing how periodic body motion and fluid interaction can amplify roll oscillations and identifying operating conditions that affect stability.
- Built sim-to-real learning pipelines using Koopman learning and differentiable reinforcement learning (RL), combining large-scale simulation with limited real-world data to learn transferable dynamics models and control policies.
- Developed online model adaptation that selectively incorporates informative streaming sensor data, enabling efficient learning across changing dynamical regimes while reducing redundant data and computation.

- Deployed real-time MPC and learned controllers on the physical robot, with low-level BLDC motor and servo control, achieving speeds up to 2 body lengths/s and 1.8–6.3 cm cross-track RMSE in closed-loop trajectory tracking.

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA — Jan 2020 – Aug 2021

Advisor: Dr. Mohammad Naghnaeian

- Designed energy-efficient planning and control strategies for UAV fleets in resource-constrained environments.
- Used MPC to enhance UAV energy efficiency, factoring in disturbances, wind effects, and morphed UAV for improved performance.

Work Experience

Engineering Intern, Thermax Ltd. – Pune, India — Oct 2017 – May 2018

- Surveyed and optimized the steel utilization in the plant.
- Developed a Python program for remote online performance analysis for the Vapour Absorption chiller plants.
- Capstone Project: Design and manufacturing of boiler panel cutting machine to replace the existing flame cutting method used in repairing boilers to avoid the development of heat-affected zones.

Engineering Intern, Elite Techno Group – Pune, India — June 2016 – June 2016

- Designed ATV on Solidworks and performed crash analysis on it.
- Manufactured the ATV and tested its performance on a road-track.

Selected Publications

Learning Koopman operators for path tracking by a fish-like robot
Kartik Loya, P. Chivkula, R. Varikuti, P. Tallapragada
(Submitted to IEEE Transactions on Robotics)

Differentiable Reinforcement Learning for Path Tracking by an Agile Fish-Like Robot
P. Chivkula, *Kartik Loya*, R. Varikuti, P. Tallapragada
(IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS))

Koopman Operator Framework for Modeling and Control of Off-Road Vehicle on Deformable Terrain
Kartik Loya, P. Tallapragada
(ASME Journal of Autonomous Vehicles and Systems (JAVS))

Agile 3-D Motion Primitives in an Underactuated Fish-like Robot
P. Chivkula, *Kartik Loya*, P. Tallapragada
(IEEE Robotics and Automation - Letters)

Parametric roll oscillations of a hydrodynamic Chaplygin sleigh
Kartik Loya, P. Tallapragada
(Nonlinear Dynamics 111 (22), 20699-20713)

Online learning of Koopman operator using streaming data from different dynamical regimes
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine 58.28, 90-95)

Koopman operator-based predictive control uses a data archive of observables
Kartik Loya, J. Buzhardt, P. Tallapragada
(ASME Letters in Dynamic Systems and Control 3 (3))

Stabilization of an Inverted Pendulum on a Nonholonomic System
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine 55 (37), 764-769)

Competitions

AI for Industry Challenge - Intrinsic × Open Robotics – Team ASIMOV — May 2026 – Aug 2026

- Ranked 21st globally and advanced to Phase 1 (Top 30 of 160+ teams) in a worldwide robotics competition focused on autonomous manipulation of deformable cables and precision connector insertion.
- Developed the manipulation solution using Intrinsic Flowstate, integrating Vision-Language-Action (VLA) models, reinforcement learning, and OpenCV-based perception for task planning, visual understanding, and robot control.

- Designed and tested reusable robot skills for perception and manipulation within the Flowstate workflow, combining learning-based approaches with task-specific robotic behaviors.

iGRADS (Graduate Research and Discovery Symposium) Mar 2026 – Apr 2026

- Advanced to the final round of Clemson's university-wide graduate research competition, presenting doctoral research to a broad, nontechnical audience.

Honors and Awards

ME Excellence Award - Ph.D. – Clemson University Apr 2026

- Recognized by the Department of Mechanical Engineering for excellence as a Ph.D. graduate student.

Graduate Travel Grant – Clemson Graduate Student Government June 2024

- Awarded competitive graduate travel funding to attend the IEEE International Conference on Robotics and Automation (ICRA 2024).

Talks and Presentations

- **IROS 2026** — Talk & Demo: "Differentiable Reinforcement Learning for Path Tracking by an Agile Fish-Like Robot.
- **Modeling, Estimation and control Conference (MECC)** — 2022: "Stabilization of an Inverted Pendulum on a Nonholonomic System"
- **Modeling, Estimation and control Conference (MECC)** — 2023: "Koopman Operator Based Predictive Control With a Data Archive of Observables"
- **Modeling, Estimation and control Conference (MECC)** — 2024: "Online learning of Koopman operator using streaming data from different dynamical regimes"
- **SIAM Conference on Applications of Dynamical Systems (DS25)** — Talk: Model-based learning of Koopman operator for complex mechanical systems (May 2025).
- **ISTVS 2025 (55th Conference)** — Talk: Data-driven Koopman control-oriented learning framework for improved autonomous off-road mobility (Oct 9, 2025).
- **ICRA 2025** — Poster: "Batch learning/updates of Koopman operators from streaming off-road vehicle data with terrain dynamics" (Summer 2025).
- **ICRA 2025** — Late Breaking Results: "Learning PID Gains for Planar Tracking of Bio-Inspired Swimming Robots Using Simplified Models"
- **Clemson VIPR Annual Meeting 2025** — Poster Presentation: "Koopman-Based Modelling and Control of Off-Road Vehicles on Deformable Terrain"
- **Clemson Graduate Student Seminar (GSR)** — Research talk: 2023 "Parametric Roll Oscillation of a Hydrodynamic Chaplygin Sleigh"
- **Clemson Graduate Student Seminar (GSR)** — Research talk: 2025 "Data-driven Koopman Control-Oriented Learning Framework for Improved Autonomous Off-road Mobility."

Coursework

Graduate Coursework: Intermediate Dynamics | Nonlinear Dynamics & Chaos | Robot Multibody Dynamics | Advanced Nonlinear Dynamics | Advance Engineering Mathematics | Linear Analysis | Control Systems Engineering | Advanced Nonlinear control | Random Processes and Application | Applied Optimal Control | Networked Dynamical systems | Advanced Estimation | Control of Cyber-Physical Systems | Autonomy: Science & Systems | Reinforcement Learning | Advanced Machine Learning | Theory of Elasticity